

Curvature-Bounded Compact Objects and the Lower Mass Gap in CPTG

Carter L Glass Jr

Abstract

Curvature Polarization Transport Gravity, or CPTG, extends compact-object physics by replacing singular collapse with gradient-bounded curvature structure. In this framework, horizon formation depends not only on enclosed mass but also on the local organization of spatial field gradients. The result is a finite-core compact-object model with direct implications for the lower mass gap, apparent compact-object masses, failed supernovae, primordial remnant survival, microlensing, kilonova structure, gravitational-wave echoes, and rare solar-system transit signatures. The central prediction is that several phenomena usually treated as separate astrophysical puzzles may be linked by one geometric mechanism: high spatial curvature gradients activate a bounded polarization response instead of producing a singular endpoint.

1 Observational Motivation

The lower mass gap is an important testing ground for compact-object theory. In the conventional picture, neutron stars occupy the low-mass side of the compact-object sequence, while stellar-mass black holes generally appear above a larger collapse threshold. Between these regimes lies a historically sparse interval separating the heaviest neutron stars from the lightest ordinary black-hole candidates.

Gravitational-wave astronomy has made this interval increasingly important. GW190814 contained a secondary compact object with a reported mass between two and one-half and two and two-thirds solar masses, placing it near the boundary between massive neutron stars and low-mass black holes [1]. GW230529 involved a compact object reported in the lower mass-gap range, with public LVK summaries describing one component between one and one-fifth and two solar masses and the other slightly more than twice as massive [2]. These detections create an observational domain where finite-core, mass-gap, and nonstandard compact-object models can be tested.

Failed-supernova and vanishing-source observations provide a second domain. N6946-BH1 remains one of the best-known failed-supernova candidates, with a luminous red-supergiant progenitor followed by dramatic optical fading [8, 9]. The

VASCO project searches historical and modern sky surveys for vanishing and appearing sources, including exceptional astrophysical transients [10]. M31-2014-DS1 in Andromeda was reported as a massive star that brightened in the infrared and then faded without a normal supernova, consistent with direct-collapse interpretation [11].

CPTG [12] links these observational domains through a single compact-object mechanism. If horizon closure depends on local curvature-gradient structure, then quiet stellar collapse, mass-gap objects, merger anomalies, and hidden compact remnants are not independent anomalies. They become different expressions of the same geometric threshold.

2 The CPTG Curvature Invariant

The compact-object mechanism begins with a quasi-static curvature invariant containing both field strength and spatial field variation:

$$C = g_j g_j + \kappa^2 (\partial_i g_j) (\partial_i g_j). \quad (1)$$

The first term represents local field strength. The second term represents spatial variation of the field, weighted by the characteristic gradient length. Both terms carry the same dimensions:

$$[C] = [g_j g_j] = [\kappa^2 (\partial_i g_j) (\partial_i g_j)] = \text{acceleration}^2. \quad (2)$$

A dimensionless curvature-response variable is introduced by normalizing against the CPTG acceleration scale:

$$\chi = \frac{C}{a_\star^2}. \quad (3)$$

The gradient-only contribution is

$$\chi_{\text{grad}} = \frac{\kappa^2 (\partial_i g_j) (\partial_i g_j)}{a_\star^2}. \quad (4)$$

With this normalization, the polarization sector can be written as

$$\mathcal{L}_{\text{pol}} = \beta a_\star^2 \chi^{1/3}. \quad (5)$$

Here, the normalization isolates the scale dependence in powers of the acceleration scale while leaving the fractional curvature response in the dimensionless variable.

The variational response of this sector to the field gradient is

$$\frac{\partial \mathcal{L}_{\text{pol}}}{\partial(\partial_i g_j)} = \frac{2}{3} \beta \chi^{-2/3} \kappa^2 (\partial_i g_j). \quad (6)$$

When the gradient term dominates, the normalized invariant follows the scaling

$$\chi \approx \chi_{\text{grad}} \propto (\partial_i g_j)(\partial_i g_j). \quad (7)$$

The high-gradient limit then gives

$$\lim_{\partial_i g_j \rightarrow \infty} \frac{\partial \mathcal{L}_{\text{pol}}}{\partial(\partial_i g_j)} \propto \lim_{\partial_i g_j \rightarrow \infty} (\partial_i g_j)^{-1/3} = 0. \quad (8)$$

This limit is the mathematical center of the compact-object model. Infinite gradients do not produce an infinite polarization force. The variational response softens, allowing the compact interior to approach a bounded geometric state rather than a singular endpoint.

3 Structural Horizon Closure

In CPTG, compact-object formation is not governed by mass alone. Horizon closure occurs when collapse creates a sufficient combination of enclosed mass, field strength, and spatial gradient structure. The dimensionless structural condition is

$$\chi_{\text{grad}} \geq \chi_{\text{crit}}. \quad (9)$$

This condition separates ordinary compact stars from gradient-triggered horizon objects. A neutron star with ordinary internal structure does not automatically become a CPTG horizon merely because its mass lies near the solar-mass range. The structural gradient must cross the critical threshold.

This distinction is essential for the lower mass gap. CPTG permits lower-mass horizon formation only in collapse states where the internal field gradients become sufficiently organized and intense. The relevant compact-object class is therefore not defined by mass alone, but by the pair

$$(M_{\text{core}}, \chi_{\text{grad}}). \quad (10)$$

The compact-object sequence is altered because a low-baryonic-mass core can become a horizon if the gradient condition is satisfied. The gap between neutron

stars and black holes is therefore transformed from a simple mass interval into a structural transition zone.

4 Finite-Core Interiors

CPTG compact objects are finite-core structures. Near the coordinate origin, the scalar potential is regularized rather than divergent:

$$\Phi(r) = \Phi_0 + \alpha_{\text{core}} r^2 + \mathcal{O}(r^4). \quad (11)$$

The coefficient fixing the local restoring curvature is determined by the finite-density interior solution. The corresponding field strength is

$$g(r) = -\nabla\Phi(r) \approx -2\alpha_{\text{core}}\mathbf{r}. \quad (12)$$

At the origin,

$$\lim_{r \rightarrow 0} g(r) = 0. \quad (13)$$

The interior therefore avoids the classical point-singularity behavior. Gravity increases toward the compact region, reaches its strongest expression near the finite-core boundary, and softens toward the center. The result is a regular compact interior with a bounded potential floor and a nonzero physical core scale.

The finite core also changes the role of the horizon. The horizon is no longer the surface surrounding a mathematical sink. It is the outer trapping boundary of a regularized compact geometry. The deepest interior remains finite, and the curvature-gradient sector supplies the mechanism preventing a zero-volume endpoint.

5 The Lower Mass Gap

The lower mass gap becomes a natural structural test of CPTG. In the conventional interpretation, the gap represents a scarcity of compact objects between the most massive neutron stars and the lightest ordinary stellar black holes. In CPTG, part of this interval can be populated by finite-core compact objects whose physical baryonic masses are lower than their apparent gravitational masses.

A representative physical core scale is

$$M_{\text{core}} \approx 1.1 M_{\odot} \text{ to } 1.4 M_{\odot}. \quad (14)$$

The apparent mass inferred from binary motion, lensing, or waveform templates can be larger if the exterior field is amplified by curvature polarization. This is written through an apparent-mass amplification factor:

$$M_{\text{apparent}} = \mathcal{A}_{\text{pol}} M_{\text{core}}. \quad (15)$$

For a representative mass-gap mapping,

$$M_{\text{core}} \approx 1.2 M_{\odot} \implies M_{\text{apparent}} \approx 3 M_{\odot} \text{ to } 5 M_{\odot}. \quad (16)$$

The corresponding polarization amplification is

$$\mathcal{A}_{\text{pol}} = \frac{M_{\text{apparent}}}{M_{\text{core}}} \approx 2.5 \text{ to } 4.2. \quad (17)$$

This apparent-mass offset gives CPTG a direct observational target. A mass-gap compact object should not behave exactly like either an ordinary neutron star or a classical Kerr black hole. It should show finite-core signatures in the late inspiral, possible tidal response, merger dynamics, and post-merger ringdown.

6 Exterior Field and Apparent Mass

The apparent-mass offset requires an exterior solution. The weak-field exterior cannot be chosen arbitrarily, because binary dynamics, lensing, timing, and gravitational waves all infer mass through the external metric. A viable CPTG compact object must therefore satisfy three simultaneous conditions.

First, the near-field exterior must reproduce the ordinary compact-object limit where general relativity is already tightly tested. Second, the intermediate field must allow curvature polarization to amplify the inferred mass without violating orbital constraints. Third, the far-field behavior must connect smoothly to the same curvature-response structure used in galaxy-scale CPTG.

A schematic near-field acceleration profile is

$$g_{\text{near}}(r) \approx \frac{GM}{r^2}. \quad (18)$$

The far-field branch is written with a structural slope index rather than a fixed exponent:

$$g_{\text{far}}(r) = g_0 \left(\frac{r_0}{r} \right)^{\nu_N}. \quad (19)$$

Here, the exponent is determined by the active CPTG structural branch:

$$\nu_N = \nu(N, \chi, \Sigma). \quad (20)$$

The familiar flat-rotation deep-field branch corresponds to

$$\nu_N = 1. \quad (21)$$

Other CPTG outer-structure branches can be represented by different values of the same exponent. This keeps the compact-object exterior from being artificially locked to one far-field slope before the full branch solution is derived.

A complete compact-object solution must interpolate between the near-field and far-field limits. The lower mass-gap prediction becomes strongest when the same exterior solution explains apparent mass, finite-core merger signatures, and the selected CPTG outer branch.

7 Evaporation Freeze-Out and Stable Remnants

The gradient cutoff also alters the possible endpoint of primordial black-hole evaporation. In semiclassical gravity, Hawking temperature rises as mass decreases:

$$T_H \propto \frac{1}{M}. \quad (22)$$

CPTG modifies this extrapolation when the horizon scale approaches the gradient length. The increasing spatial variation term changes the near-horizon thermodynamic behavior and motivates a freeze-out endpoint instead of complete evaporation.

The corresponding minimum scale is written as an order-of-magnitude relation:

$$M_{\min} \sim \frac{\kappa c^2}{G}. \quad (23)$$

If the horizon scale is fixed through the Schwarzschild radius convention,

$$r_s = \frac{2GM}{c^2}, \quad (24)$$

then the corresponding numerical factor is

$$M_{\min} \sim \frac{\kappa c^2}{2G}. \quad (25)$$

The remnant radius follows the gradient length:

$$r_{\text{core}} \sim \kappa. \quad (26)$$

In this picture, primordial compact objects do not necessarily vanish in a final evaporative burst. They can settle into stable or long-lived sub-microscopic remnants. Such remnants would be electromagnetically quiet because their physical core scale is extremely small, but they could still produce gravitational effects through lensing, orbital perturbations, or collective far-field tracking.

Primordial black holes are already constrained by microlensing, dynamics, accretion, evaporation, and gravitational-wave observations. Reviews of primordial black holes as dark-matter candidates emphasize that allowed mass windows are highly model-dependent and observationally restricted [4, 5].

8 Cosmological Accounting

A large remnant population must be compatible with cosmological density measurements. Planck measurements of the cosmic microwave background separate baryon density from cold dark matter density in the standard cosmological model, with cold dark matter contributing substantially more than ordinary baryons [3].

CPTG remnant abundance is therefore controlled by a cosmological accounting rule. Four possibilities define the viable parameter space. The remnants may constitute only a subcomponent of the dark sector. The early-universe inference of baryon and cold-dark-matter densities may change under CPTG. The remnant interior may represent a bound phase not counted as ordinary free baryonic matter after formation. Or the neutron-core language may describe the stabilized internal phase while the cosmological abundance is determined by the modified gravitational sector.

The compact-remnant proposal becomes strongest when the same population satisfies CMB abundance limits, microlensing limits, dynamical limits, and galaxy-scale CPTG requirements simultaneously.

9 Microlensing and Hidden Compact Objects

Stable remnants provide a natural microlensing test. Their electromagnetic cross-section can be extremely small, but their gravitational field can still magnify background sources. A hidden population would generate short-duration, dark-lens events with no luminous counterpart.

CPTG differs from ordinary primordial-black-hole lensing if the exterior field transitions into a polarization-dominated profile. The lensing kernel is then not

identical to a point-mass general-relativistic lens. This creates a direct observational discriminator: event duration, amplification curve, finite-source behavior, and spatial distribution must be calculated from the CPTG exterior metric.

Masked abundance is a related effect. Some compact objects cataloged as ordinary low-mass black holes could be lower-baryonic-mass finite-core objects whose exterior fields make them appear heavier. The test is not the catalog mass alone. The test is whether the object carries correlated deviations: finite-core tidal response, altered merger ejecta, or post-merger echo structure.

10 Merger Signatures

Compact-object mergers provide a direct strong-field test. A classical black hole has no material surface and, in the standard treatment, no tidal deformability. A finite-core CPTG object can possess an effective tidal response because its interior is regular rather than singular.

The predicted inequality is

$$\Lambda_{\text{CPTG}} > 0. \quad (27)$$

A nonzero effective deformability changes the late-inspiral phase. The deviation should grow most visible near merger, where the compact objects enter the high-gradient regime.

Kilonova emission offers a complementary signal. In a standard neutron-star–black-hole merger, bright ejecta require disruption outside the black-hole horizon. A finite-core mass-gap object can produce a different disruption pattern because its physical baryonic mass and apparent gravitational mass are not identical. The predicted electromagnetic signatures include neutron-rich ejecta, altered angular symmetry, and a light curve that may resemble a double-neutron-star event more than a classical black-hole–neutron-star event.

The strongest merger test combines three observables:

$$(\Lambda_{\text{eff}}, M_{\text{apparent}}, E_{\text{ejecta}}). \quad (28)$$

A mass-gap object with an overmassive gravitational appearance, nonzero tidal response, and unexpectedly rich ejecta would occupy the natural CPTG target region.

11 Gravitational-Wave Echoes

Finite-core compact objects can also produce delayed ringdown structure. Exotic compact-object models show that objects mimicking black holes during inspiral and merger may generate secondary pulses or echo-like residuals after the main ringdown if the near-horizon region is partially reflective [6]. Searches using LIGO and Virgo data have placed constraints on echo amplitudes and templates [7].

In CPTG, the reflecting structure is the regularized finite core. A characteristic echo delay is

$$\Delta t_{\text{echo}} \approx \int_{r_{\text{core}}}^{r_H} \frac{2}{c\sqrt{-g_{00}g_{11}}} dr. \quad (29)$$

The echo sequence is not arbitrary. It is controlled by the core radius, the horizon-scale geometry, and the effective reflectivity of the gradient-bounded interior. The expected signal is a coherent sequence of delayed, high-frequency residuals following the main burst.

The same finite-core parameters controlling the echo delay also influence tidal deformability. Echoes, tidal phase shifts, and apparent-mass offsets therefore form a linked strong-field prediction.

12 Failed Supernovae and Vanishing Stars

Quiet stellar collapse is another CPTG target. In standard failed-supernova scenarios, the core collapses directly into a black hole while the optical supernova is weak or absent. Observed candidates such as N6946-BH1 and M31-2014-DS1 show that disappearing massive stars are astrophysically relevant phenomena [8, 9, 11].

CPTG expands the mechanism by lowering the effective collapse threshold when the gradient sector activates. The critical process is

$$\chi_{\text{grad}} \rightarrow \chi_{\text{crit}}. \quad (30)$$

As a stellar core contracts, the spatial gradient contribution grows. When the critical gradient condition is reached, the polarization response can close a horizon before a normal outward shock successfully ejects the envelope. The observational result is a quiet or failed explosion, infrared brightening from disturbed outer material, and subsequent optical disappearance.

Stable primordial remnants create a secondary channel. A remnant captured by a star could settle into the interior, increase local gradients, and modify the thermal balance. This channel predicts slow photospheric fading, weak electromagnetic output, and a possible absence of ordinary supernova ejecta. The rate

depends on remnant abundance, capture cross-section, stellar structure, and the transport suppression factor.

13 Solar-System and Seismic Traces

If stable remnants exist in significant abundance, rare solar-system encounters become possible. These objects would not behave like asteroids because their physical radius is controlled by the gradient scale rather than by ordinary material density. They would not create ordinary impact craters, atmospheric fireballs, or broad ejecta blankets.

A simple separation estimate follows from the local mass density and remnant mass:

$$d_{\text{sep}} \approx \left(\frac{M_{\text{remnant}}}{\rho_{\text{local}}} \right)^{1/3}. \quad (31)$$

A representative scaling is

$$d_{\text{sep}} \approx 15 \text{ AU} \left(\frac{M_{\text{remnant}}}{10^{19} \text{ g}} \right)^{1/3}. \quad (32)$$

The observable signals are gravitational and mechanical. A planetary transit would create a linear seismic Mach structure rather than a normal tectonic earthquake. The event would align with a hyper-velocity straight-line path through the body and would lack an ordinary fault-plane origin.

On the Moon or Mercury, a transit could leave narrow through-bore structures, high-pressure mineral phases, or unusual entry-exit symmetry. In the Sun, a transit or capture event could produce helioseismic ring waves or subtle core perturbations. These signatures are rare-event predictions and require population synthesis before they can be used quantitatively.

14 Information Storage and Horizon Smoothness

The finite-core model also changes the information problem. If evaporation freezes out at a finite core scale, the horizon area does not collapse to zero. A minimum surface scale remains:

$$A_{\text{min}} \sim \kappa^2. \quad (33)$$

The transport sector provides a geometric redistribution mechanism:

$$\mathcal{L}_{\text{trans}} \propto - (S^\alpha \Sigma \nabla_\alpha \Phi)^2. \quad (34)$$

The transport vector is normalized as

$$S^\alpha S_\alpha = \pm 1. \quad (35)$$

The suppression factor is bounded by

$$0 \leq \Sigma \leq 1. \quad (36)$$

In the high-gradient interior, transport suppression increases:

$$\Sigma \rightarrow 0 \quad \text{as} \quad \chi_{\text{grad}} \rightarrow \chi_{\text{crit}} \quad \text{or} \quad \chi \rightarrow \infty. \quad (37)$$

The transport vector organizes the directional flow of curvature and information, while the suppression factor limits propagation in the high-gradient interior. Infalling structure is redistributed across the finite core boundary rather than being driven into a zero-volume singularity.

The same gradient softening that regularizes the core also prevents the horizon from becoming a discontinuous high-energy wall. The high-gradient variational response tends toward zero instead of infinity, preserving a smooth crossing condition at the effective horizon.

15 Falsifiable Predictions

CPTG compact-object physics leads to a linked observational program:

- mass-gap compact objects should exist as a continuous structural population rather than as isolated anomalies;
- some apparent black holes in the lower mass range should show finite-core signatures;
- apparent gravitational mass can exceed physical baryonic core mass through exterior polarization;
- mass-gap mergers can show nonzero effective tidal deformability;
- mass-gap mergers can produce unexpectedly neutron-rich or symmetric kilonova ejecta;
- post-merger ringdowns can contain coherent echo-like residuals tied to the finite-core scale;

- primordial remnants can survive if the CPTG evaporation endpoint freezes out;
- remnant populations can produce dark microlensing events with nonstandard lensing kernels;
- quiet stellar collapse and vanishing-star events can occur at altered structural thresholds;
- rare remnant transits can create linear seismic or planetary through-bore signatures.

These predictions are mutually constraining. The same parameters controlling the compact-object core must also control apparent mass, tidal deformability, echo delay, remnant survival, and lensing behavior. That linkage makes the framework falsifiable: one cannot freely tune each observational channel independently.

16 Conclusion

CPTG replaces singular compact-object endpoints with finite, gradient-bounded curvature structures. This single change reorganizes the physics of black-hole formation, lower mass-gap objects, primordial evaporation, merger signals, vanishing stars, and hidden compact remnants.

The most important prediction is not simply that more compact objects exist. The deeper prediction is that mass, horizon closure, and external gravitational appearance are structurally separated. A compact object may have a low physical core mass, a stronger apparent gravitational mass, finite-core tidal behavior, and post-merger echo structure. Those signatures should appear together if the same gradient-bounded mechanism is responsible.

The next stage is a quantitative compact-object program: exterior metric construction, horizon-closure criteria, evaporation thermodynamics, population synthesis, microlensing kernels, kilonova calculations, and waveform templates. With those pieces in place, the lower mass gap becomes a direct observational test of CPTG rather than a purely interpretive anomaly.

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